

Spatiotemporal coherence of MHT in the Atlantic Ocean

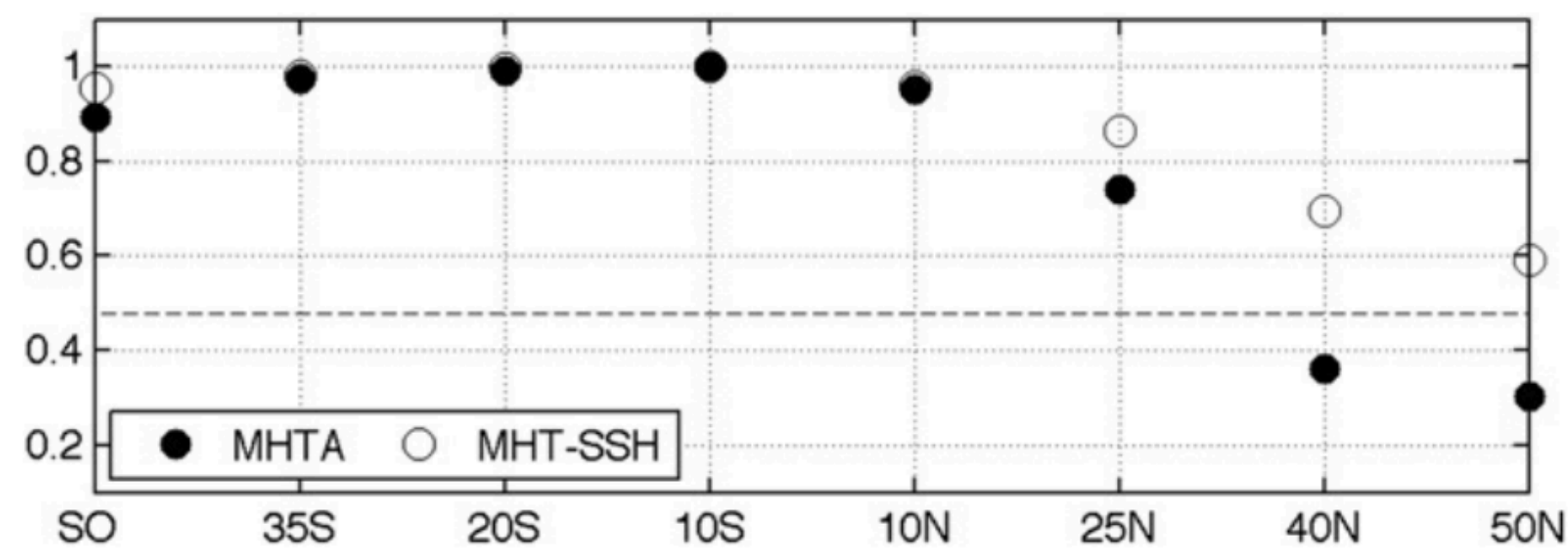
1. Event : breakdown/split of coherence around 2009

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Literature says:

Kelly et al. (2014):

- Coherence south of 40°N up to SO
- Timeperiod: 1993-2011

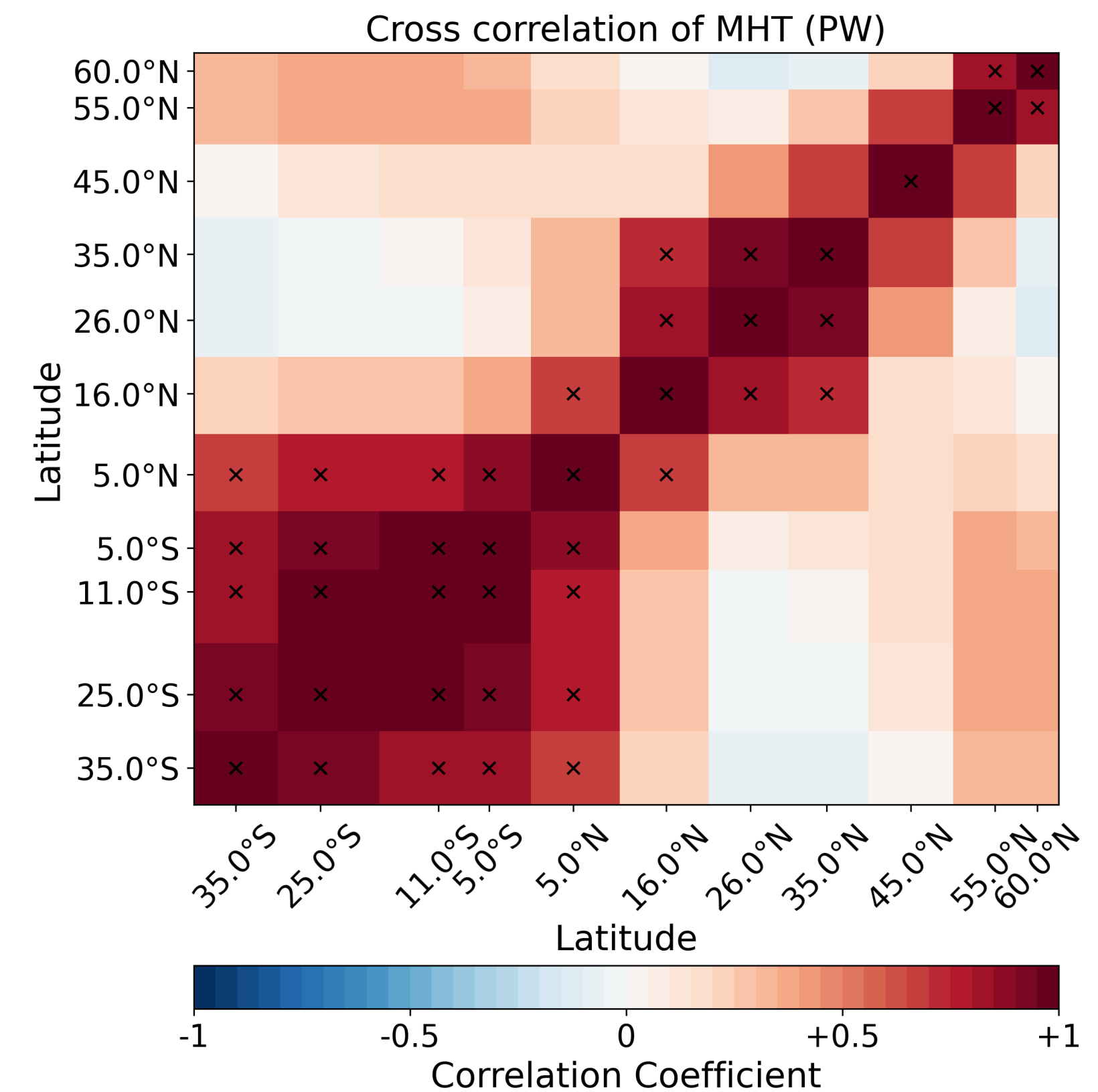


When is this split from coherence up to 40°N and 2 covarying blocks happening and what is causing it?

Calafat et al. (2025):

- 2 Covarying blocks:
- 35-16°N and 5°N-35°S
- Timeperiod: 2004-2020

Cross correlation of MHT anomalies from Calafat dataset:



Idea:

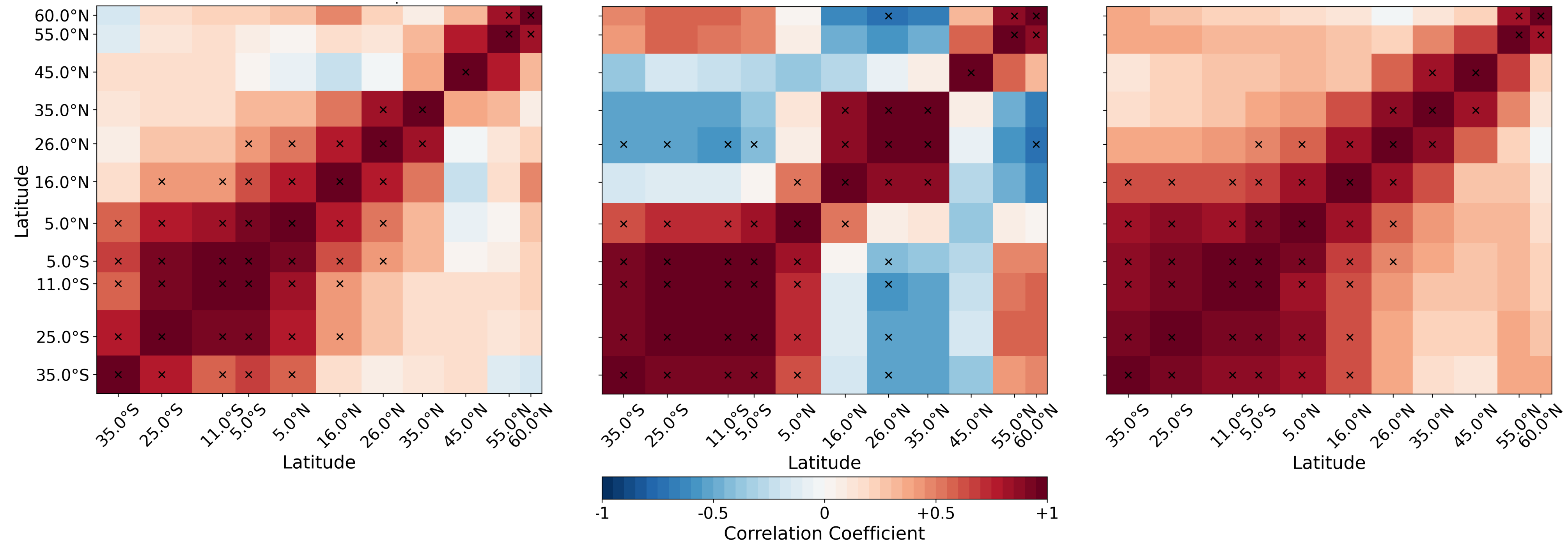
- **split timeseries into before and after split year**
 - **plot gif sliding the split year through the timeseries**
- This led me to distinguish 3 time periods**

Split whole timeseries into 3 periods:

Pre 2009

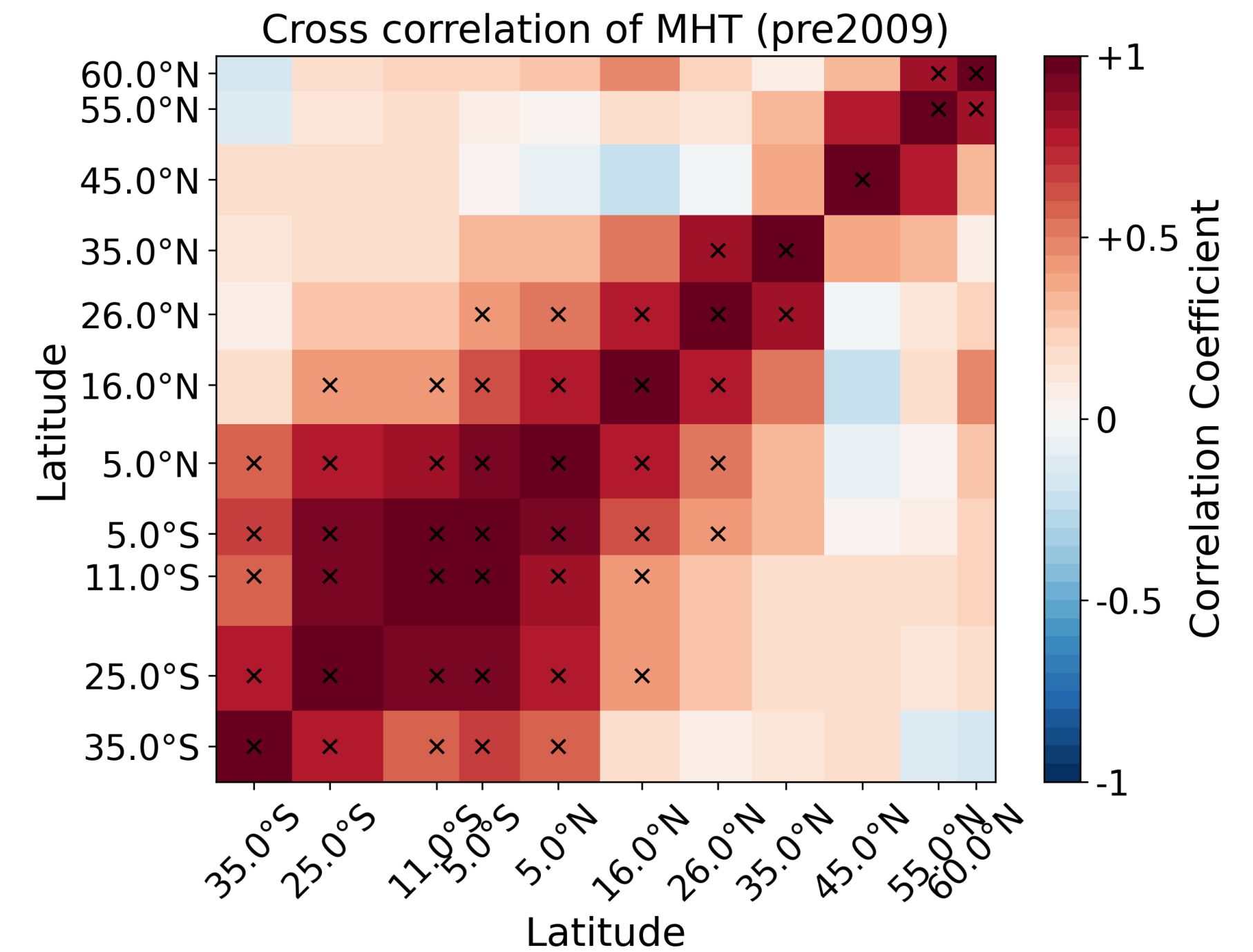
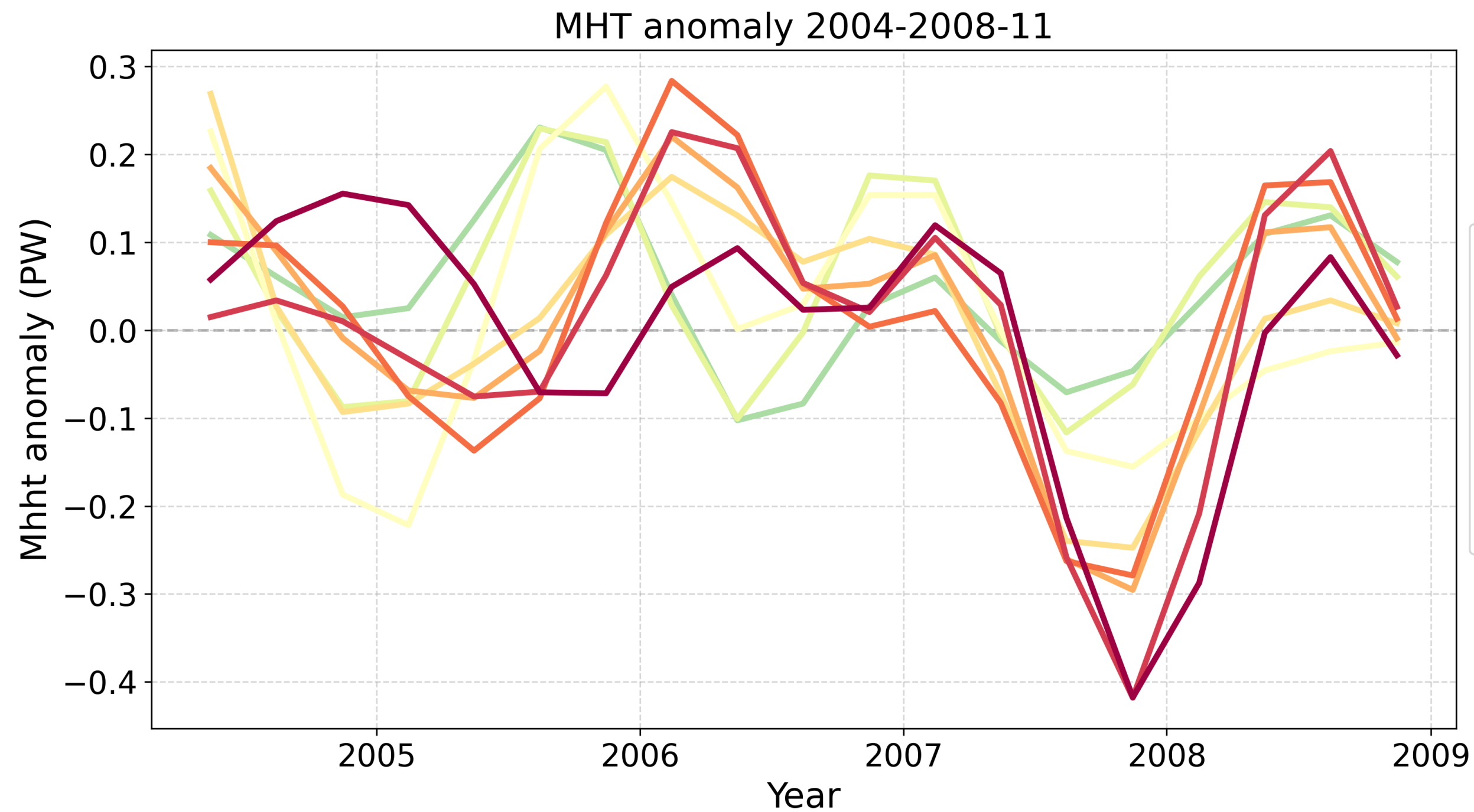
2009-2013

Post 2013



- Before 2009: high correlation of MHT anomalies between latitudes throughout Atlantic south of 35°N
- Then pattern shifts and moves southwards, splits into two boxes around 2009
- Two covarying boxes stays until 2013

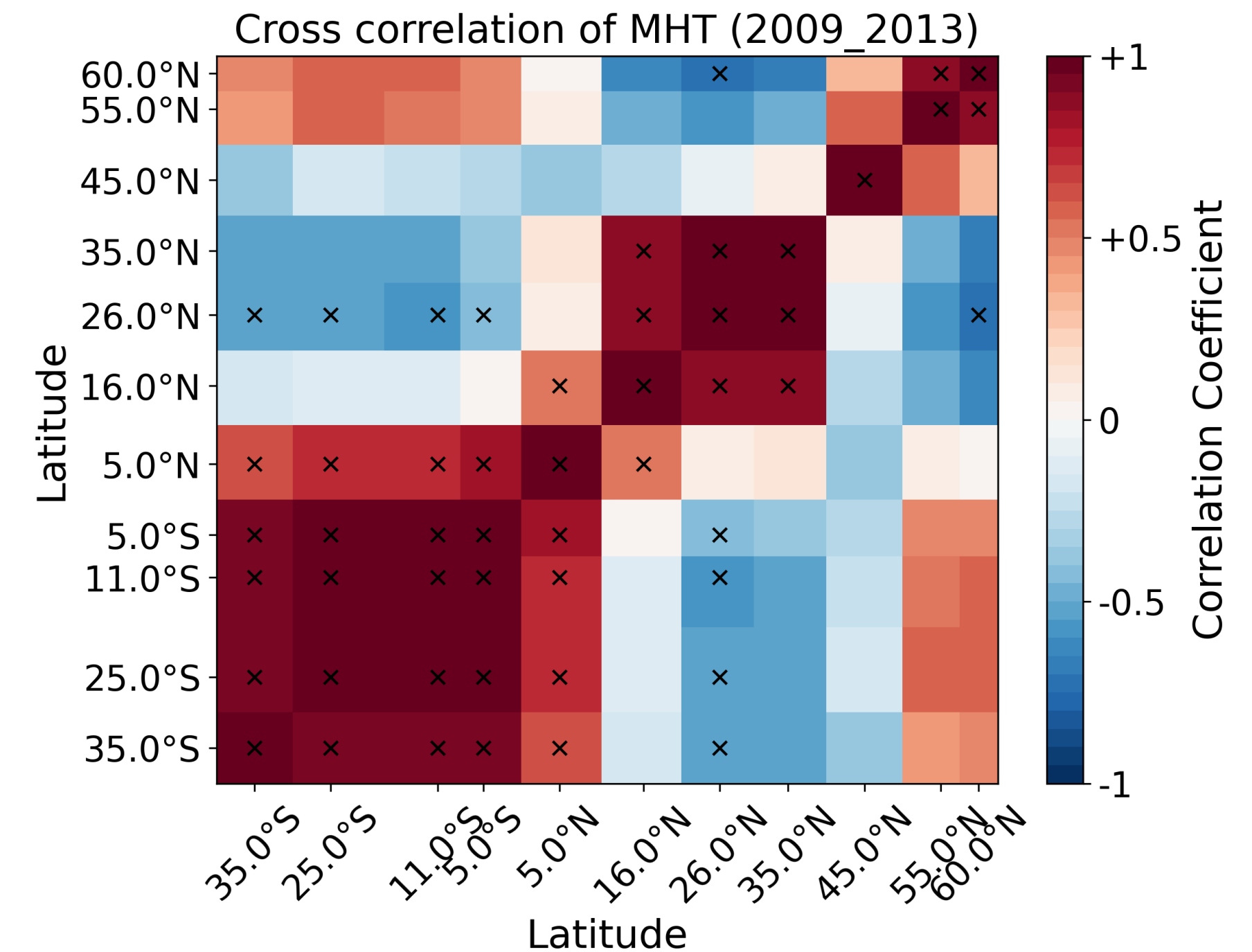
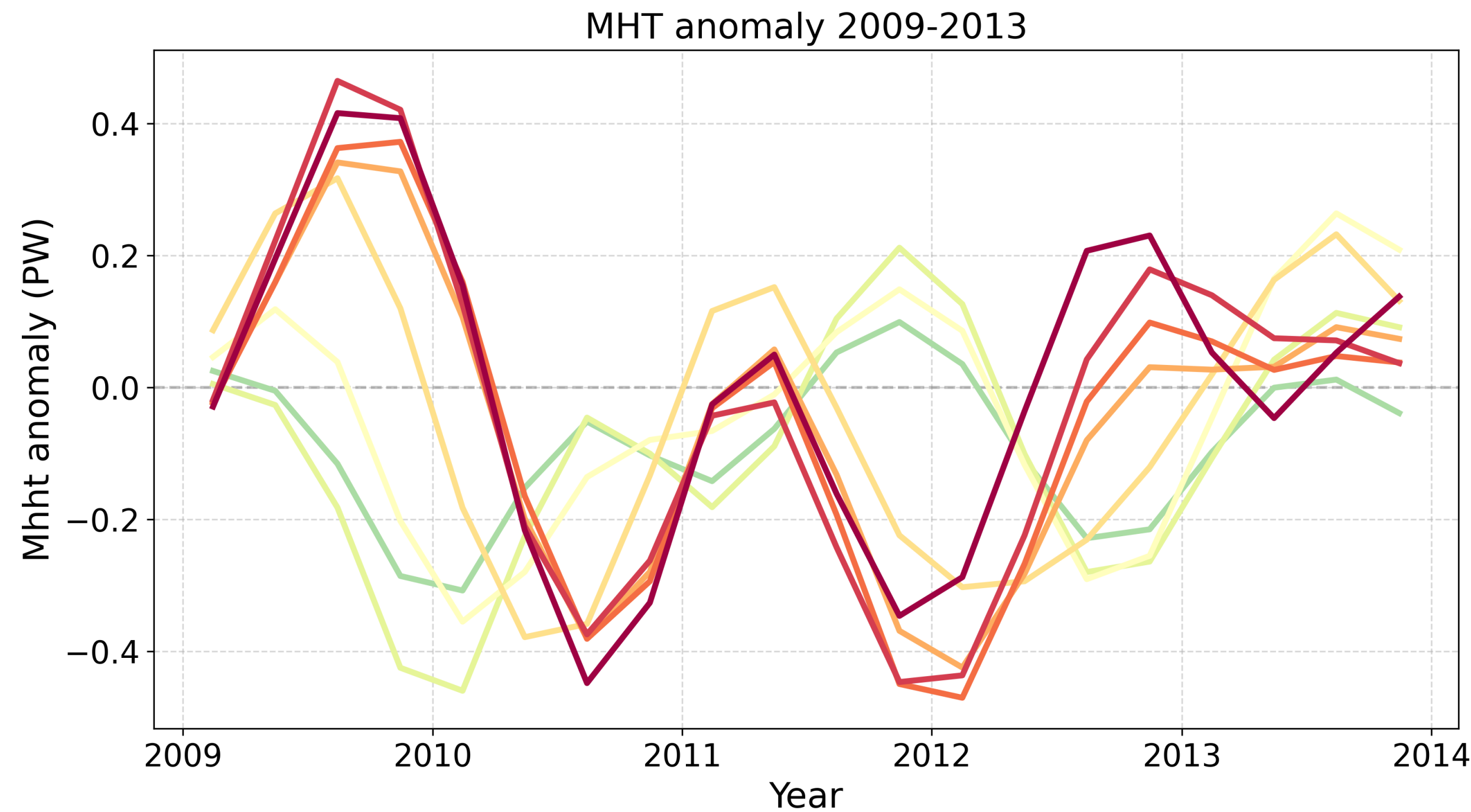
MHT Coherence before 2009



Focus on latitudes south of 45°N: **35°N - 35°S**

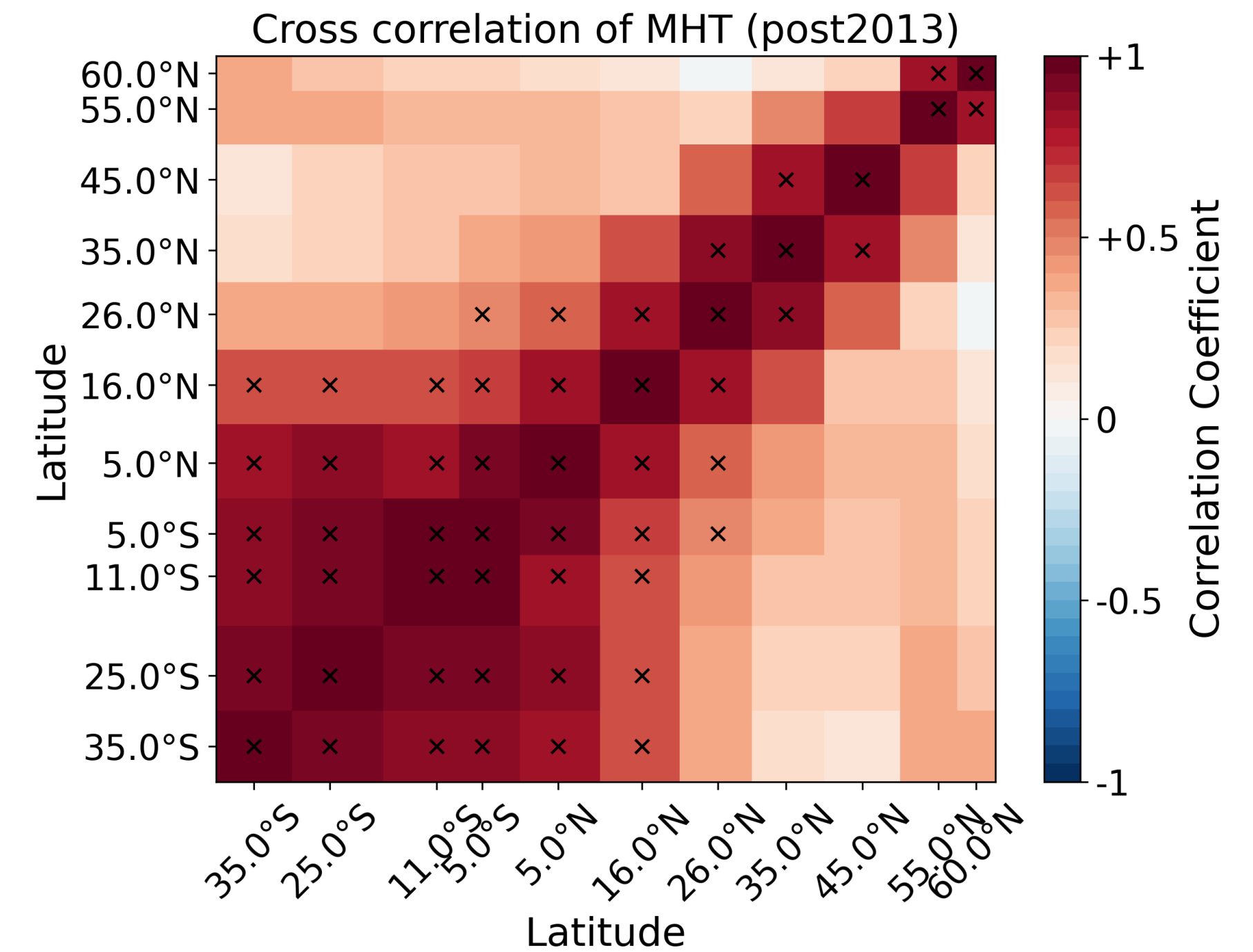
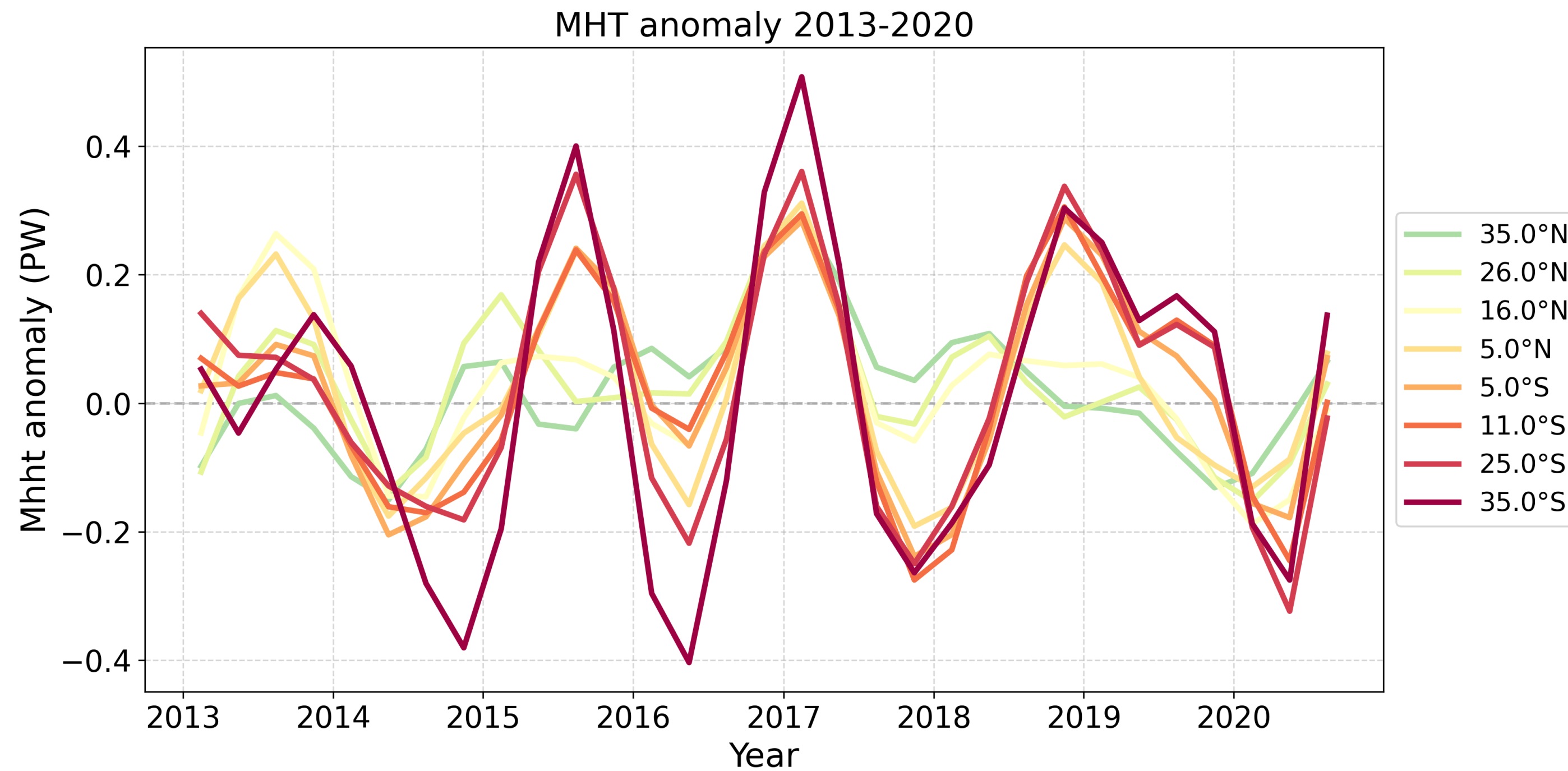
- Signal mostly **in phase** (especially from 2007)

MHT Coherence / correlation break 2009 - 2013



- Here we see signal goes from in phase to **anti-correlated**
- split into 2 covarying boxes:
 - 35 - 16°N (Subtropical block (**ST**))
 - 5°N - 35°S (South Atlantic block (**SA**))

MHT Coherence after 2013



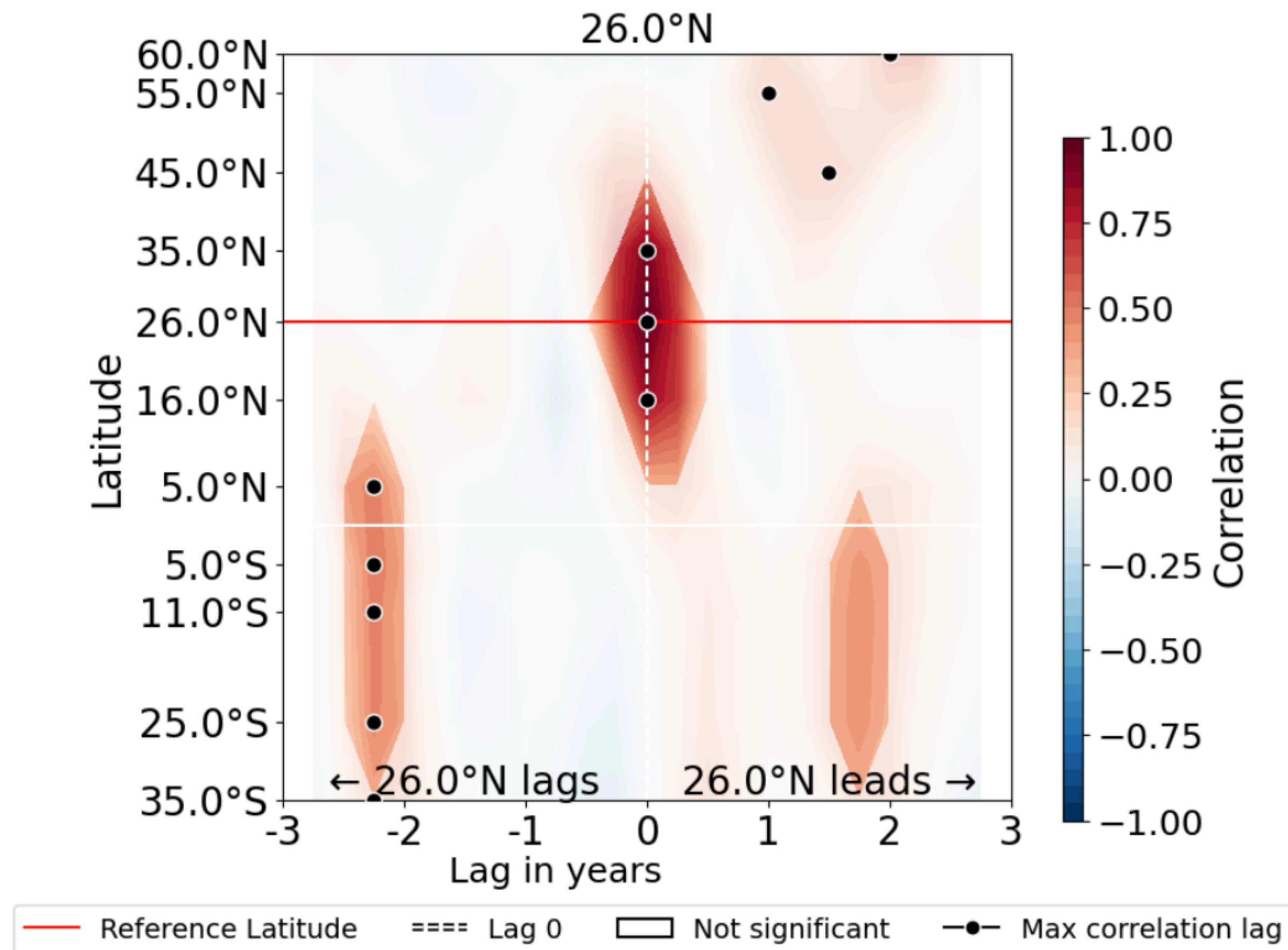
- Again mostly **in phase signal** and **amplitude**, northern latitudes tend to have lower amplitudes

Quantification of coherence break / split:

	Pre 2009	2009-2013	Post 2013
Spread across block 35°N - 35°S	0.08 PW	0.14 PW	0.09 PW
Corr between ST and SA block	0.54	-0.31	0.63
Corr within ST	0.80	0.90	0.85
Corr within SA	0.79	0.86	0.90

- Spread is the mean STD over that latitude block
- 2009-2013 shows anti correlation between the ST (35 - 16°N) and SA (5°N - 35°S) block

Lead / lag analysis of MHT anomalies



From lead / lag analysis only 26°N shows significant relationship:

- 35 - 16°N Band clearly visible
 - Lags 5°N - 25°S by ~2.5 yrs (maximum correlation lag)
- signal propagation south to north
- But also leads by ~2 yrs, less pronounced

Next steps / open questions:

Find propagation in the signal, do a max lag correlation plot

- Quantify the break down with in terms of how much heat and what timescales?

What caused the coherence break and split into two boxes?

- 2008/2009 record low NAO → AMOC & MHT drop
- Connect NAO - air sea heat flux F_S - MHT (HTC)